

Pyramid Problems · Loops — English Worksheet

Topic: Loops · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
repeat 4 times:
  place block down
  move forward
```

What does the agent build? How many blocks?


```
repeat 6 times:
  place block down
  move up by 1
```

Is this a wall lying flat, or a tower standing up? How tall?


```
repeat 5 times:
  move forward
```

Does the agent place any blocks? Where does it end up?

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, rewrite it, then explain why the original was wrong.

Bug A — The agent is supposed to use a loop to place a row of **5 blocks**.

```
place block down
move forward
place block down
move forward
place block down
move forward
place block down
move forward
place block down
```

Write the fixed code using a loop:

Why was the original code messy? Why does your loop work?

Bug B — The agent is supposed to build a tower **5 blocks tall**, with one block at each height.

```
repeat 5 times:
  place block down
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a row of 6 blocks, all in a straight line on the ground.

```
repeat 6 times:  
  place block down  
  move forward  
  turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world and use a loop to build a **20-block straight wall**. Then build a **U-shaped wall** (three sides).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.